Strategic Roadmap: Quantum-Native ROI

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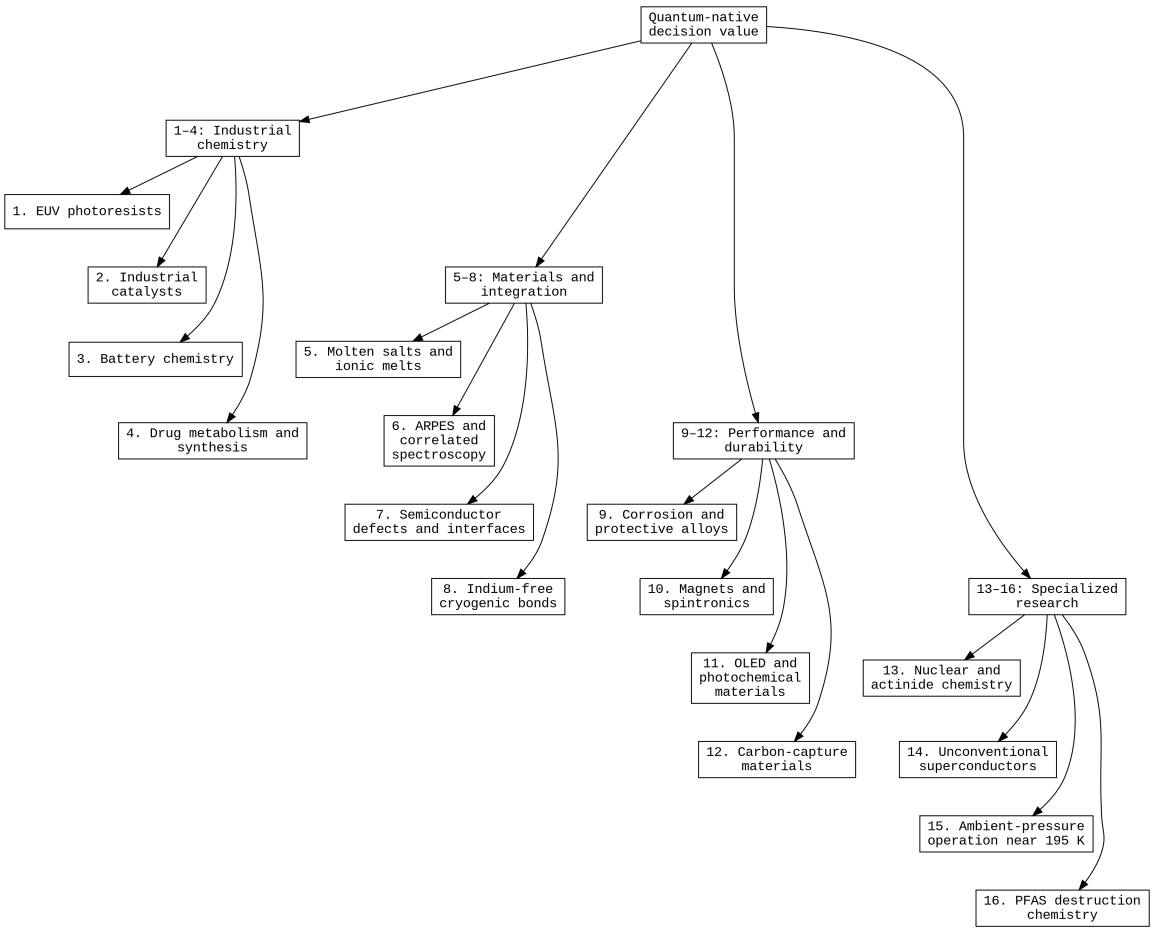
October 16, 2026

Quantum-native calculations create economic value when better predictions change a material, process, or development decision. Here, we prioritize EUV photoresists, industrial catalysts, battery chemistry, and selected pharmaceutical mechanisms because they connect difficult quantum observables to repeated, expensive industrial decisions. Molten salts, correlated spectroscopy, semiconductor defects, and cryogenic interconnects extend this strategy into targeted engineering programs.

The objective is to fund a sustainable computing program through validated decisions, paid research, and deployed improvements. Computational difficulty identifies a technical opportunity; captured decision value determines the investment case.

1. Opportunity tree

The branches preserve the overall priority order. Each application receives a detailed assessment in the ranked chart



2. Planning framework

We rank opportunities using five criteria: commercial leverage, feasibility of the selected quantum kernel, the classical baseline, experimental validation requirements, and access to an accountable decision owner. This is a strategic allocation of research effort. Application utility, realistic resources, and total cost enter the same assessment. [1]

Quantum-native describes a target whose governing model is quantum mechanical, including electronic Hamiltonians, quantum spin models, and open quantum systems. CPUs and non-QPU accelerators already solve many such models numerically. A proposed QPU route adds an explicit encoding, algorithm, accuracy target, and

resource estimate. We track the physical model, execution platform, predictive accuracy, and commercial outcome separately.

The conventions below apply throughout the report, allowing the application sections to focus on decisions and execution.

Planning element	Working convention	How it guides investment
Priority order	We prioritize the strongest combination of decision leverage, executable scope, validation access, and value capture.	Reorder projects when partner access, resources, or decision deadlines change.
Evidence	Published demonstrations and resource studies establish the specific model, method, and scale tested.	Use the evidence labels below to select the next technical milestone.
Classical baseline	Compare the QPU-enabled workflow with the strongest practical classical-plus-experimental alternative at the decision date.	Fund a quantum kernel where its incremental predictive value exceeds its complete delivery cost.
Research and qualification windows	Budget active research and downstream qualification separately; schedule hardware readiness, queues, and full quantum execution through the critical path.	Use variable discovery windows and dated engineering milestones rather than fixed discovery promises.
Financial scenarios	The coverage budget, industrial scales, delivery probabilities, capture fractions, and discount rate are planning inputs in constant US dollars.	Replace these inputs with partner economics and architecture-specific costs before committing capital.
Attribution	Benefits enter the QPU contribution ledger when quantum execution changes the selected action relative to the baseline.	Link prediction, validation, deployment, and captured value.
Planning element	Working convention	How it guides investment

Portfolio accounting	Charge shared costs once and assign each commercial improvement to one outcome.	Treat spectroscopy as a shared capability and the 195 K conductor target as part of superconductivity.

Evidence labels

Label	Evidence class	Information retained
H	Execution on physical quantum hardware	Tested model, physical qubits, reference method, accuracy, and scale.
R	Published quantum resource estimate	Logical or physical qubits, gate units, repetitions, runtime assumptions, and excluded steps.
S	Algorithm, simulator, or emulated demonstration	Classically executed operations and the scope of validation.
P	Proposed application-specific QPU route	The mapping, predictive gap, and resources to establish in a feasibility study.

These labels describe different forms of evidence. Hardware execution establishes an implemented protocol; economic advantage requires the full decision and cost comparison.

Classical baseline ratings

Rating	Competitive landscape	Project response
High	Strong classical methods address much of the workflow or the relevant embedded instances.	Select a documented residual error or cost bottleneck at the required decision accuracy.
Moderate-high	Difficult correlated instances offer a research route alongside substantial classical competition.	Benchmark the selected observable, environment, and deadline directly.
Unresolved	Model-to-material prediction, synthesis, or the comparative method remains an open research question.	Fund bounded hypothesis tests and update the investment case at each milestone.

3. Ranked opportunity chart

Rank	Application and decision value	QPU kernel	Evidence	Classical baseline and competition
1	EUV photoresist chemistry. Select formulations, exposure dose, and defectivity controls.	Absorption, correlated photoemission, and selected reactions through time-domain simulation.	R; distinct absorption and continuum-dyna mics resource estimates, with experimental resist-chemistry support. [2, 3]	Moderate-high. Classical excited-state chemistry, electron transport, and empirical process models cover substantial portions of the workflow.
2	Industrial catalysts. Improve selectivity, feedstock utilization, energy use, and catalyst life.	Embedded electronic energies and reaction barriers through qubitized quantum phase estimation.	R; Ru/CO2 and nitrogen-fixation instances. [4, 5]	Moderate-high. DFT, coupled cluster, DMRG, and embedding provide strong references for selecting difficult correlated steps.
3	Battery chemistry. Choose composition, redox pathways, and degradation controls.	Energy differences and core response through QPE, XAS, and RIXS algorithms.	R; model spectra and material-property estimates. [6, 7, 8]	High for broad screening; moderate-high for selected spectra. Classical thermodynamic s and learned potentials support the large configuration search.

4	Drug metabolism and selective synthesis. Resolve metal-site mechanisms and select synthesis routes.	Spin-state ordering and reaction energies through embedded QPE.	R; a P450 study compares classical and quantum resources. [9]	High for general drug workflows; moderate-high for selected metal sites. Prioritize electronic uncertainties that alter a development or process decision.
5	Molten salts and ionic melts. Select recovery, redox, corrosion, and electrolyte operating conditions.	Embedded correlated energies through sample-based quantum diagonalization or prospective QPE, coupled to classical liquid sampling.	H/P; FLiBe fragment execution identifies embedding as a major accuracy bottleneck. [10, 11]	High for bulk sampling; moderate-high for electronic corrections. MD and free-energy methods establish the thermodynamic baseline.
6	ARPES and correlated spectroscopy. Resolve electronic models that guide material selection.	Spectral functions, core response, and correlations through fermionic dynamics and QPE-based response.	H/R; a 54-qubit ARPES-related protocol benchmark uses a free-fermion model. [12, 7]	High for weak correlation; moderate-high for selected correlated response. Compare with many-body solvers and measured spectra.
7	Semiconductor defects and interfaces. Select charge states, spin states, and device-relevant structures.	Localized electronic states through quantum defect embedding, VQE, and subspace methods.	H; diamond NV and SiC divacancy models. [13]	High for small embedded spaces. Classical impurity and excited-state solvers address many existing test instances.

8	Indium-free cryogenic bonds. Improve assembly yield, electrical performance, and reliability.	Selected phase and interface electronic kernels within classical alloy and mechanics workflows.	P; sub-kelvin interconnect studies provide experimental controls. [14, 15, 16]	High. CALPHAD, DFT, mechanics, and experiments establish the screening workflow; target a specific electronic limitation.
9	Corrosion and protective alloys. Extend component life and reduce maintenance burdens.	Reaction and oxidation energies through embedded chemistry or first-quantized QPE.	R; Mg/Nb model studies provide logical resource estimates. [17]	High for multiscale screening; moderate-high for selected chemistry. Environmental modeling and sampling accompany electronic accuracy.
10	Magnets and spintronics. Select exchange interactions, transport regimes, and device materials.	Spin correlations and response through spin evolution and open-system algorithms.	H/R for model kernels; funded materials programs establish application interest. [18, 19, 20]	Moderate-high. Tensor networks, Monte Carlo, and classical transport offer instance-specific alternatives.
11	OLED and photochemical materials. Select emission, color, and stability characteristics.	Excited-state energies and spin transitions through excited-state VQE and dynamics.	H; small active-space TADF calculations. [21]	High. Classical excited-state chemistry and targeted experiments provide strong references; device stability adds a separate validation stage.

12	Carbon-capture materials. Select adsorption, regeneration, and sorbent-life tradeoffs.	Local adsorption and reactive chemistry through embedding and fragment solvers.	S; emulated Al-fumarate/CO 2 workflow. [22]	High. Classical electronic screening, Monte Carlo, and MD address many adsorption tasks.
13	Nuclear-energy and actinide chemistry. Support reaction, separations, and storage decisions.	Relativistic correlated states through quantum moments and prospective QPE.	H; trapped-ion actinide-model demonstrati on. [23]	Moderate-high. Specialized relativistic and multireference classical methods define the comparison.
14	Unconventional superconductors. Select models and candidate material families.	Correlations and finite-temperatu re pairing through Hamiltonian and thermal simulation.	R for selected model calculations. [24]	Unresolved at the product level. Evaluate the route from model accuracy to synthesis and useful conductor properties.
15	Ambient-press ure operation near 195 K. Pursue a defined conductor objective within superconductivi ty.	Pairing, stability, and response coupled to synthesis and conductor qualification.	P, supported by related model-resource studies and experimental temperature /pre ssure benchmarks. [24, 25, 26, 27]	Unresolved. Select testable hypotheses linking a numerical prediction to a stable, useful material.
16	PFAS destruction chemistry. Select bond-breaking mechanisms and treatment routes.	Correlated electronic pathways through prospective electronic-Hamil tonian QPE.	P; an implemente d classical incremental-FCI study provides a strong baseline. [28]	High. Modern correlation algorithms and parallel cloud computation already advance the relevant frontier.

EUV leads because its molecular observables feed a high-value manufacturing process repeated across many wafers. Catalysts compete closely through recurring feedstock and energy savings. Batteries and selected metal-site chemistry follow through high-volume production and expensive development decisions. For an accessible pilot, a paid spectroscopy, salt-property, or interconnect study may reach a decision sooner than these larger programs.

4. EUV: connect molecular accuracy to fab decisions

The direct quantum calculation produces an absorption strength, ionization pathway, electron-emission distribution, or reaction observable. Classical electron-cascade, reaction, and film models mediate the indirect connection to resist blur, defectivity, and wafer yield. The research task is to improve the molecular input that currently limits a formulation decision. [2]

We prioritize a resist family and observable together. Polymer length alone is a weak selection criterion because the difficult physics may reside in a local excitation, correlated decay, or dissociation pathway. Metal-containing resists expand the scope beyond polymers; tin-oxo cage experiments connect absorption and fragmentation directly to resist chemistry. [3]

Project stage	Concrete deliverable	Decision enabled
Define the resist question	Named resist family, exposure conditions, measured observable, and acceptable error.	Select the calculation that distinguishes candidate formulations.
Establish the classical baseline	Excited-state chemistry, transport inputs, and empirical process predictions under matched conditions.	Identify the error that obstructs the formulation choice.
Validate the electronic prediction	Blind absorption, emission, or reaction comparison against measurement.	Accept or reject the molecular input for the process model.
Propagate through the process workflow	Updated cascade, reaction, and film predictions with sensitivity analysis.	Choose a resist or process change.

Qualify the manufacturing outcome	Measured defectivity, dose, yield, and process-window performance.	Deploy the improvement and attribute its economic contribution.

Absorption screening is the first resource-estimated entry point. Full photoemission targets a broader continuum-dynamics problem with a much larger workload, so its investment case rests on the additional formulation value it delivers. [2]

5. Classical baseline and benchmark discipline

Classical methods determine both the cost of the alternative and the accuracy needed to change a decision. Improved embedding, selected configuration interaction, tensor networks, Monte Carlo, and learned potentials continually shift this comparison. Lee et al.'s empirical assessment found insufficient evidence for a generic exponential ground-state chemistry advantage across chemical space, reinforcing the need for instance-specific benchmarks. [29]

Classical development	Published evidence	Strategic response
Massively parallel electronic correlation	A PFAS study distributes incremental-FCI work across up to one million simultaneous cloud vCPUs. [28]	Compare the relevant bond-breaking observable, actual total cost, and deadline with the implemented classical workflow.
Atomistic foundation models	MACE-MP-0 demonstrates broad atomistic modeling and application-specific fine-tuning. [30]	Use GPU-based models for configuration search and dynamics; identify the electronic reference-data gap for a proposed QPU contribution.
Learned-potential salt thermodynamics	A LiCl study predicts a melting point of $880 \pm 18 \ \ K$ against an experimental 883 K. [11]	Benchmark melting-point and free-energy decisions against validated classical sampling workflows.

Classical initial-state construction	Quantum preparation methods use classical wavefunction descriptions as inputs to energy estimation. [31]	Measure the joint benefit: a stronger classical solver improves the baseline and supplies a better QPU starting state.

A learned potential accelerates sampling of its learned energy surface. The reference data and experimental checks determine the physical validity of that surface. This creates a targeted hybrid route: supply selected high-value electronic reference data, then use classical models for the large-scale search.

We refresh each benchmark quarterly and whenever a material algorithmic, hardware, or experimental result changes the comparison.

Benchmark field	Required content	Decision purpose
Owner and deadline	Decision owner, candidate actions, and date.	Ensure that useful information reaches someone able to act.
Physical target	Material, geometry, environment, and observable.	Compare the same physical question across methods.
Decision precision	Acceptable observable error and confidence.	Set accuracy from the action threshold rather than a universal energy tolerance.
Model specification	Hamiltonian, basis, active space, and approximations.	Separate model accuracy from solver accuracy.
Alternatives	Strong classical methods and a targeted experimental route.	Establish the current cost, accuracy, and delivery baseline.
Error budget	Model, basis, embedding, solver, sampling, and instrument uncertainties.	Direct effort toward the uncertainty that limits the decision.
Complete workload	Preparation, compilation, shots, decoding, mitigation, analysis, retries, and staff time.	Cost the delivered answer.
Validation	Blind comparison and a documented acceptance threshold.	Establish whether the prediction changes the action.

Commercial terms	Deployment path, captured share, and payment timing.	Translate the improvement into program cash flow.

Retain classical workflows that meet the required decision accuracy at lower total cost, and direct QPU work toward the remaining predictive gaps. This allocation supports useful quantum-native services while building a stronger case for quantum hardware.

6. Time to decision and time to cash

Time to decision ends when the owner receives enough validated information to select an action. Time to cash ends when a payment arrives or a deployed change produces a saving. A paid interpretation study often reaches cash before a new material completes product qualification.

The schedule starts quantum execution when the hardware, data, and application workflow are ready. Full execution, analysis, and decision validation then determine the result date. Product qualification, contracting, adoption, and payment terms determine the cash date. Concurrent tasks share calendar time; the dependency schedule identifies the critical path.

Timing component	Schedule input	Deliverable
Hardware readiness	Required logical resources, fidelity, runtime, and access.	A workload-ready execution platform.
Data and workflow preparation	Geometry, parameters, model selection, references, and implementation.	A reproducible calculation and comparison.
Quantum execution	Repetitions, sweeps, state preparation, queues, and retries.	The complete observable dataset.
Decision validation	Experimental or blinded predictive tests.	Evidence supporting the selected action.
Qualification	Reliability, scale-up, integration, and application requirements.	A process or product ready for deployment.
Commercial realization	Contract, adoption, production ramp, and payment terms.	Captured cash or operating savings.

Program planning allowances

The windows below use the planning framework in Section 2. Research windows describe active campaigns with relevant people, data, samples, and facilities. Qualification starts after a usable result; hardware waiting and quantum runtime enter the critical-path schedule separately.

Rank	Program	Research/ decision window	Qualification window	Primary cash-flow route
1	EUV	6-18 months	18-48 months for formulation/fab qualification.	Paid formulation study or qualified process change.
2	Catalysts	6-18 months	18-48 months for pilot/process integration.	Paid mechanism study or deployed process saving.
3	Batteries	6-18 months	24-60 months for cell/process qualification.	Paid material decision, licensing, or production value.
4	Pharma/synthesis	6-18 months	12-36 months for a process route; model therapeutic development separately.	Research contract or qualified synthesis route.
5	Molten salts	6-18 months	18-60 months for a pilot formulation; reactor deployment follows its program schedule.	Paid property package or operating-formu lation value.

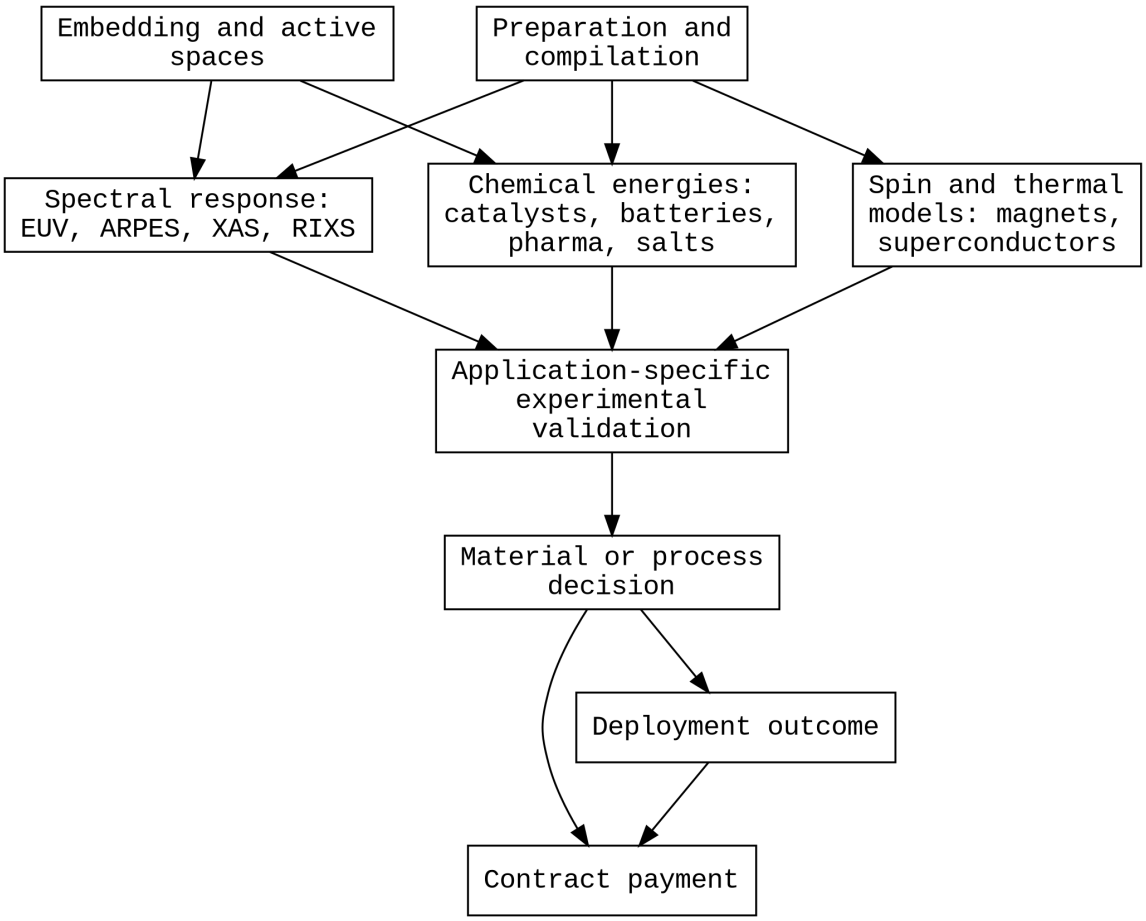
6	ARPES/spectroscopy	3-12 months	0-12 months for a paid interpretation service; product qualification follows separately.	Contracted experimental decision.
7	Semiconductor defects	6-18 months	18-48 months for a representative device/process.	Qualified device material or paid model study.
8	Cryogenic bonds	6-12 months	12-36 months for assembly/reliability qualification.	Improved assembly yield or reduced rework and retests.
9	Corrosion/alloys	6-18 months	24-60 months for component-relevant qualification.	Material selection followed by lifetime value.
10	Magnets/spintronics	6-18 months	18-48 months for device/material qualification.	Device prototype, contract, or production change.
11	OLED/photochemistry	6-18 months	18-48 months for stability and device qualification.	Emitter licensing or product integration.
12	Carbon capture	6-18 months	24-60 months for cycling and pilot qualification.	Sorbent contract or operating saving.
13	Nuclear chemistry	6-24 months	Application- and program-specific qualification.	Contracted research or qualified program decision.

14	Unconventional superconductors	6-24 months for a bounded model decision.	Variable discovery window, followed by useful-conducto r qualification.	Research contract, followed by conductor value after successful discovery and qualification.
15	Ambient-pressure 195 K target	6-24 months to test a specified hypothesis.	Variable discovery window, followed by conductor qualification.	Research contract, followed by application value from a useful material.
16	PFAS treatment chemistry	6-18 months	18-48 months for treatment/pilot qualification.	Paid route decision or deployed treatment saving.

Date classical-service income and contracted research payments separately from later QPU-derived application gains. That distinction identifies the funding available during hardware and application development.

7. Shared capabilities and portfolio value

Embedding, preparation, compilation, and response algorithms recur across applications. We build shared capabilities around specified workloads, then measure the reduction in gates, sampling, integration effort, or staff time. Application-specific models and validation determine how much of that saving transfers. [4, 10, 13, 31]



The branches represent computational dependencies and validation routes. A shared algorithm directly reduces a kernel's resource requirement; its contribution to industrial value is mediated by the validated application decision.

Shared capability	Main destinations by rank	Transferable benefit	Application-specific work
Hamiltonian construction, embedding, and active spaces	1-5, 7-9, 11-13, 16.	Smaller difficult subproblems and reusable data interfaces.	Convergence, polarization, environmental effects, and fragmentation bias.
Initial-state preparation and energy filtering	2-5, 9, 13-16 where the selected energy/thermal algorithm uses them.	Fewer failed projections and lower repeated estimation costs.	Ground-state or thermal-state requirements, preparation quality, and construction cost.

Factorization, block encoding, and compilation	1-5, 9, 13, with method-specific extensions.	Lower logical gate counts or shorter circuits.	Operator norms, basis, precision, hardware mapping, and overhead.
Response functions and spectral reconstruction	1, 3, 6, 10, 11.	Shared correlation measurement and analysis infrastructure.	Probe operators, core/valence structure, broadening, temperature, and instrument response.
Ensemble sampling and free-energy integration	3, 5, 8, 9, 12, 14, 15.	Reusable classical sampling and selected electronic corrections.	Phase, timescale, composition, rare events, and thermodynamic target.
Resource estimates, error budgets, and benchmark automation	1-16.	Comparable decisions and less duplicated engineering.	Per-project acceptance criteria and measured costs.

If an electronic kernel occupies 20% of total workflow cost and a method halves its cost, the direct workflow saving is 10% before integration. A gate reduction that advances hardware readiness offers a different benefit, measured through the delivery schedule.

The FLiBe benchmark makes the accuracy distinction concrete: fragment solutions agree closely with their classical references while fragmentation introduces much larger binding-energy errors. Improving the reusable solver alone leaves the dominant physical prediction error unresolved. [10]

Portfolio accounting

Assign shared infrastructure and development costs once, then add each application's integration and validation cost. Count a battery decision supported by both RIXS and energy calculations as one commercial improvement. Allocate the 195 K conductor objective within the superconductivity program.

Model common hardware delays, preparation bottlenecks, and embedding failures as shared scenario events. Their effects propagate across dependent projects. Correlation

changes portfolio loss exposure and funding runway even when individual project expectations remain unchanged.

8. Execution platforms for quantum-native models

Numerical workflow	Executing hardware	Quantum-native target	QPU development route
DFT, TDDFT, coupled cluster, DMRG, selected CI, and incremental FCI.	CPUs; GPUs where supported by the implementation.	Electronic structure, excitations, and bond breaking under method-specific approximations.	Replace a selected difficult electronic solver while retaining classical model construction and validation.
GW, DMFT, exact diagonalization, tensor networks, and impurity solvers.	CPUs or non-QPU accelerators.	Correlated spectral functions and electronic response.	Execute selected embedded Hamiltonians or real-time correlations on a QPU.
Spin-Hamiltonian diagonalization, tensor-network dynamics, and master-equation integration.	CPUs/GPUs.	Quantum spin states, correlations, and open-system dynamics.	Use digital spin simulation or an executable open-system algorithm for the target observable.
Classical quantum-circuit simulation.	CPUs/GPUs.	Numerical representation of a quantum algorithm.	Validate circuits and estimators before physical QPU execution.

Hardware VQE or quantum sampling with a classical outer calculation.	QPU + CPU/GPU.	An active-space state or correlated subproblem.	Extend the demonstrated protocol to decision-relevant accuracy and competitive scale.
Fault-tolerant QPE and Hamiltonian simulation.	Logical QPU plus classical control, decoding, and application computing.	Electronic energies and dynamics at specified precision.	Translate logical resources into an architecture, physical qubits, runtime, and full-workload cost.

P450, battery spectra, spin defects, and PFAS provide concrete examples of these execution distinctions. [6, 9, 13, 28]

9. Published resource

The table retains each study's resource units and workload scope. T gates, Toffoli gates, physical qubits, and logical qubits measure different requirements. Repetitions and observable coverage convert a circuit estimate into a full calculation.

Calculation	Published estimate or demonstration	Program implication
EUV absorption, model monomer IMePh.	About 200 logical qubits, 10^9 non-Clifford gates per circuit, and 10^3 shots for the smallest instance. [2]	Approximately 10^{12} gate invocations across those shots; the study uses a screening target around 10% precision for strong absorbers.
EUV photoemission, model monomer.	Several thousand logical qubits, at least 10^{14} non-Clifford gates per circuit, and about 10^4 shots. [2]	At least approximately 10^{18} gate invocations across those shots, making this a substantially larger workload than absorption.

Battery XAS, Li4Mn2O cluster.	100 logical qubits and fewer than $4 \times 10^{\wedge}8\ \ T$ gates per circuit; 18 spatial orbitals and 22 active electrons. [6]	A defined spectroscopy target validated on a classical simulator; include sampling and the selected spectral coverage.
Battery RIXS, Mn-oxygen-dimer cluster.	414 logical qubits and $2 \times 10^{\wedge}10$ Toffoli gates per shot for the 20-spatial-orbital instance; about 2,000 shots for the reconstruction target. [7]	Approximately $4 \times 10^{\wedge}13$ Toffoli invocations at one incoming frequency, followed by any additional frequencies, polarizations, or structures.
Cytochrome P450, largest studied models.	About 5 million physical qubits, $7.8 \times 10^{\wedge}9$ Toffoli gates, and under 100 hours in the specified surface-code configuration. [9]	Cost a molecular electronic-structure workload and then add the surrounding development workflow.
Mg/Nb corrosion models.	2,292-38,598 logical qubits and approximately $1.04 \times 10^{\wedge}13 - 1.962 \times 10^{\wedge}16\ \ T$ gates for one QPE shot, excluding initial-state preparation. [17]	Compare the full model with useful embedded alternatives and include preparation in the cost.
ARPES-related spectral functions.	A 27-site, one-dimensional free-fermion model executed with 54 physical qubits on Quantinuum H2 in a 2026 preprint. [12]	Establishes protocol execution for a classically solvable benchmark; the next target is a matched correlated-response comparison.
PFAS electronic structure, classical benchmark.	Up to one million simultaneous cloud vCPUs; PFOA calculations correlate 150 electrons in 330 orbitals using incremental FCI. [28]	A demanding quantum-native classical reference for a proposed QPU comparison.

Full-workload costing

Resource field	Treatment in the cost model
Preparation and success probability	Include state construction, overlap/filtering, thermal preparation where needed, and restarts.
Circuit gates and depth	Preserve gate type and account for compilation, parallel execution, and hardware throughput.
Shots and observable coverage	Include frequencies, spectral points, polarizations, structures, compositions, and temperatures.
Logical-to-physical mapping	Specify error correction, physical error rates, factories, interconnects, and decoding.
Classical work	Include embedding, diagonalization, fitting, data handling, ensemble generation, and reference calculations.
Annual throughput	Include queues, calibration, maintenance, uptime, and utilization.
Predictive accuracy	Combine the error budget at the decision observable, preserving dependencies among error sources.

Published utility valuations

Benchmark	Published utility and resource basis	Economic interpretation
Two steps in a homogeneous nitrogen-fixation/cyanate catalytic cycle.	$200,000 estimated utility, 139,000 QPU-hours, and about 400,000 CPU-hours for the compared DMRG approach. [5]	Utility divided by QPU time is about $1.44 per QPU-hour, a gross coverage ceiling before other costs. CPU-hours and QPU-hours require separate pricing and throughput models.
Metal-insulator-transition materials search.	Roughly $400M model-based research-search utility, alongside severe runtime obstacles. [19]	A scenario valuation of a research search, distinct from revenue for one computation.

$Ca_3Co_2O_6$ nonequilibrium magnetism.	Roughly $2M per-material utility proxy based on scientific funding. [19]	A research-value measure with a different economic basis from product margin or avoided experimental expenditure.

These studies make the investment test explicit: fit a useful calculation inside the cost, runtime, and validation budget of the decision it supports.

10. Investment and coverage model

We use an annual coverage target of $20M to size the program: $8M of annual capital allocation plus $12M of operating cash cost. Two ledgers connect this target to the work.

The program cash ledger records receipts and expenditures that fund construction, staff, operations, and research. It includes net classical-service income and contracted research payments. The QPU contribution ledger records the incremental decision value attributable to quantum execution relative to the best practical alternative. Classical quantum-native services provide a funding route during hardware development, with proceeds assigned to their actual source.

Coverage budget: planning inputs

Cost component	Planning basis	Annual-equivalent amount
Initial system, fit-out, and build effort.	$40M allocated over five years.	$8M
Engineering and scientific staff.	20 staff at $250,000 fully loaded per year.	$5M
Facilities, maintenance, and classical compute.	Annual cash allocation.	$3M
External application software and contracted workflow services.	Annual cash allocation, excluding the listed staff payroll.	$2M
Experimental validation.	Annual cash allocation.	$2M
Annual coverage target	$8M capital allocation + $12M operating cash cost.	$20M

This model defines the coverage target. The engineering budget for a selected physical or logical processor supplies the architecture-specific cost input.

Break-even logic

Let (G) be the annual incremental operating value after validated deployment, measured against the classical-plus-experimental baseline. Let (p) be the probability of reaching that deployed state within the modeled horizon, (f) the share captured by the program, and (C) the annual-equivalent program cost. For fixed conditional value and capture:

$$B = pfG\,,\ \ R_{\text{annual}} = \frac{B - C}{C},\ \ G_{\text{break-even}} = \frac{C}{pf}$$

At $(p = 0.5)$, $(f = 0.5)$, and $(C = \$20\,M)$ break-even requires \$80M of conditional annual operating value. At full delivery and full capture, the requirement is \$20M. Use contribution margin and avoided expenditure to measure value; total market revenue supplies context. When delivery, capture, and value vary together, calculate expected receipts from their joint scenarios.

Industrial scales: independent planning scenarios for an \$80M annual gain

Application	Annual activity or cost basis	Improvement producing $(G = \$80\,M)$	Causal link
EUV fabrication.	\$10B of net contribution-equivalent output at 100% yield, with fixed product mix and incremental margin.	+0.8 percentage points of absolute yield.	Improved molecular prediction → resist/process change → measured yield gain.
Industrial catalysis.	\$2B of addressable feedstock, energy, and process-loss costs.	4% cost reduction.	Improved selectivity or barrier prediction → deployed process change → lower operating cost.

Battery production.	10M packs per year.	$8 net gain per pack.	Chemistry selection → cell/pack/manufacturing qualification → realized unit gain.
Pharmaceutical development or synthesis.	$1B of genuinely addressable annual costs.	8% cost reduction.	Mechanism or route prediction → eliminated expenditure or improved process economics. Weight avoided failures by their probabilities.
Corrosion and reliability.	$10B of addressable maintenance, replacement, and downtime costs.	0.8% cost reduction.	Validated material/process change → lower failure or maintenance burden.
Spectroscopy service.	$2M of genuinely avoidable expenditure per campaign.	40 campaigns avoided annually.	Validated prediction → replaced campaign expenditure. Paid interpretation follows its contracted value instead.

In the EUV scenario, $(10\,\mathrm{B} \times 0.008 = 80\,\mathrm{M})$. The 0.008 increase is 0.8 percentage points, and the contribution-equivalent denominator converts yield into operating value. The process change mediates the link between a molecular calculation and the financial result.

Discounting and deployment timing

Place receipts and expenditures at their occurrence dates and discount them to a common date, following life-cycle cost practice. [32] For scenario (s), let $(B(s,t))$ be captured benefit, $(C(s,t))$ operating cash cost, $(K(s,t))$ additional capital expenditure, and (K_0) initial capital:

$$NPV\,s = -K_0 + \sum t=1^{T} \frac{B(s,t) - C(s,t) - K(s,t)}{(1+r)^t}, E[NPV] = \sum_s \pi_s NPV_s.$$

Scenario probabilities (π_s) sum to one and include common delivery events once. Classical competition changes the future incremental value (G); deployment success determines whether receipts begin.

Five-year cash-flow scenario input	Value and treatment
Initial capital	$40M at year 0.
Operating cash cost	$12M at each year-end for years 1-5, committed in all outcomes.
Conditional annual deployed value	$(G = \$160\,M)$.
Capture fraction	$(f = 50)$, yielding $80M annual receipts after successful deployment.
Delivery probability	$(p = 50\%)$, yielding $40M expected annual benefit after receipts begin.
Discount rate	10% real, with constant-dollar cash flows.
Horizon	End of year 5; zero terminal value.
Other cash flows	No renewal capital, taxes, or inflation in this scenario.
Capital accounting	Charge the initial $40M once; omit the $8M annual allocation from the cash-flow ledger.

Receipt delay	First cash year	PV expected benefit	PV program cost	Expected NPV	Break-even conditional annual (G)
0 years	1	$151.6M	$85.5M	+$66.1M	$90.2M
1 year	2	$115.3M	$85.5M	+$29.8M	$118.7M
3 years	4	$52.2M	$85.5M	-$33.3M	$262.3M

Immediate receipts require about $90.2M of annual conditional value for discounted break-even, compared with $80M in the annual coverage screen. Upfront capital and discounted receipts explain the difference. Later deployment leaves fewer earning years inside the fixed horizon; extending the horizon requires the associated operating, staffing, renewal, and terminal-value entries.

Outcome exposure and classical value erosion

First cash year	NPV after successful deployment	NPV after delivery failure	Planning success probability	Success probability required for nonnegative expected NPV
1	+$217.8M	-$85.5M	50%	28.2%
2	+$145.0M	-$85.5M	50%	37.1%
4	+$18.8M	-$85.5M	50%	82.0%

Expected NPV guides allocation; the failed-campaign loss determines the funding exposure borne by the organization. Stage commitments around conditional costs, cancellation opportunities, and common hardware dependencies to keep that exposure aligned with runway.

For year-4 first cash, halving incremental annual value from $160M to $80M reduces expected NPV to approximately $-\$59.4\,M$. Refreshing the classical benchmark updates this value directly. Keep the discount rate, technical probabilities, and cash flows separate so that explicit technical failure is counted once.

11. Material-characterization simulations

Characterization provides a concrete route from a quantum observable to an experimental decision. The QPU computes a selected response; classical accelerators retain fitting, instrument modeling, inference, and integration with the experiment.

Experimental technique	Quantum observable or kernel	Commercial decision	Present and proposed execution path
Angle-resolved photoemission spectroscopy, ARPES.	Occupied momentum-resolved spectral function $\left(A\left(k,\omega\right)\right)$; add photoemission matrix elements and instrument response to predict intensity.	Distinguish electronic reconstructions, correlation models, and useful doping/interface regimes.	Classical many-body calculations; QPU model spectral-function demonstrations and larger quantum time evolution. [12]

Resonant inelastic X-ray scattering, RIXS.	Core-excited intermediate propagators and valence-excitation amplitudes.	Interpret oxygen-redox and degradation signatures in battery materials.	Classical cluster/response solvers; QSP/QPE-based algorithm with published resources. [7]
X-ray absorption spectroscopy, XAS/XANES.	Dipole response, core excitations, and spectral features.	Assign oxidation states and local electronic structures.	Classical calculations; resource-estimated time-domain quantum response. [6]
X-ray photoelectron spectroscopy and EUV photoemission.	Electron removal, ionization, and continuum dynamics.	Identify chemical/surface states and parameterize resist electron cascades.	Classical spectral modeling; quantum correlation and continuum-dynamics routes. [2, 12]
Inelastic neutron scattering, INS.	Dynamical spin structure factor $\left(S(q,\omega)\right)$.	Determine exchange interactions and magnetic excitation models.	Classical spin solvers and tensor networks; hardware-executable spin evolution and correlation measurement. [20]
Optical absorption, fluorescence, and intersystem crossing.	Excitation energies, transition strengths, spin-state ordering, and dynamics.	Select emitters, photoactive molecules, and optical defects.	Classical excited-state chemistry; small-model QPU demonstrations and dynamical extensions. [13, 21]
Transport and metal-insulator-tran sition characterization.	Current correlations, transport coefficients, and open-system response.	Assess switching materials and device operating regimes.	Classical transport/impurity methods; open-system quantum algorithms with resource analysis. [19]

For ARPES, the deliverable is a paid prediction, an improved material decision, or a validated experimental saving. Specify energy and momentum resolution, temperature, precision, sample count, and state preparation before costing a full spectral map. This connects QPU + XPU execution to the actual characterization campaign.

12. Cryogenic interconnect alloy and interface discovery

An indium-free interconnect program starts with the bonding process, cryogenic operating condition, and electrical role. Superconducting microwave links, normal-metal thermalization links, and mechanical spacers require different properties. Interface chemistry, oxides, geometry, stress, and thermal transport determine assembled performance alongside the bulk material. Published flip-chip work identifies interface engineering as a substantive design task. [14, 15]

Candidate or control	Reason to include it	Validation target
Sn and SnAg-based bumps.	Established alternatives provide a practical indium-free reference; SnAg microbumps have sub-kelvin characterization. [14]	Critical current or resistance at the required temperature and field, relevant microwave loss, and thermal-cycle survival.
Multi-component Sn-rich alloys with barrier/passivation stacks.	A tin-alloy superconducting-bond patent defines a candidate composition and interface space for testing. [16]	Phase stability, diffusion, intermetallic formation, oxidation, bonding temperature, yield, and cryogenic loss.
Direct Cu or superconducting direct-bond routes.	Alternative assembly processes establish the competing process economics; Cu hybrid bonds have cryogenic characterization. [14]	Electrical/thermal resistance, assembly tolerances, reliability, compatible process temperature, and suitability for the intended electrical role.
Improved indium stack.	A matched control separates the benefit of bulk-alloy replacement from better interface processing. [15]	The same geometry, operating conditions, and acceptance criteria used for every candidate.

The QPU project targets a specific phase, defect, interface, or electronic-correlation error that blocks the classical screen. CALPHAD, DFT, mechanics, and experiments retain the alloy-screening and package-engineering roles. Monetize improved assembly yield, avoided cryogenic retests, and reduced rework.

Package-yield sensitivity: independent-link planning model

For $(N = 10{,}000)$ critical links with independent, identical survival probability (y), the all-good package probability is $(Y = y^N)$.

Individual-link survival input	Calculated all-good package yield
99.990%	36.79%
99.999%	90.48%

This sensitivity identifies the economic lever: a small improvement at each link compounds across many critical connections. Use correlated-failure and redundancy models when the actual package contains common defects or spare links.

13. Ambient-pressure superconductors operating near 195 K

Normal-pressure dry-ice sublimation occurs near 194.69 K, or -78.46 °C. The conductor objective is operation near 195 K at ambient pressure, with a critical temperature above the operating point and useful current and field margins. [25]

Reference or objective	Temperature and pressure	Strategic meaning
Sulfur hydride associated with H3S.	Reported (T_c) around 203 K at approximately 150 GPa. [26]	Establishes superconductivity in this temperature range under high pressure.
Pressure-quenched $HgBa2Ca2Cu3O8+\delta$.	A 2026 study reports a 151 K ambient-pressure transition after high-pressure preparation. [27]	Supports metastable-state retention as a search direction while setting a lower-temperature experimental reference.

Engineering objective.	Ambient-pressure operation near 195 K, with (T_c) above that operating temperature.	Requires current-density, field, lifetime, thermal-cycle, and fabrication qualification.

We organize discovery around chemically tuned correlated compounds, pressure-stabilized phases with credible recovery routes, strain/interface engineering, and electron-phonon systems with specified predictive gaps. Leveraging Hamiltonian studies as a starting point [24], candidate selection hinges on a multifaceted evaluation of finite-temperature phase behavior, pairing evidence, and material stability, in addition to ground-state energies.

The application determines ROI: refrigeration and maintenance savings, useful conductor or magnet performance, or an enabled product. Include pressure preparation, containment where required, AC losses, stabilization, conductor fabrication, and operating margin in the system comparison. A useful conductor delivers value through its own application, including applications outside qubit hardware.

14. Molten salts and related ionic melts

FLiBe tritium chemistry provides a concrete hardware entry point. Das et al.'s QPU + XPU fragment benchmark reports accurate fragment solutions alongside much larger fragmentation-induced binding-energy errors. We therefore prioritize embedding quality and the reconstructed chemical observable alongside solver development. [10]

Family	Representative system	Calculation target	Value connection
Fluoride fusion and reactor salts.	FLiBe, LiF-BeF2; FLiNaK, LiF-NaF-KF.	Tritium speciation/release, redox, impurity coordination, and corrosion energetics.	Fuel recovery and material compatibility; selected FLiBe fragment energies have QPU execution. [10, 33]
Chloride heat-transfer and reactor salts.	MgCl2-KCl-NaCl and fuel-bearing chloride families.	Hydrolysis, impurities, redox, solubility, and salt/alloy reactions.	Purification, high-temperature heat transport, and component life; fuel salts add actinide and radiation requirements. [34, 35]

Nitrate/nitrite thermal-storage salts.	NaNO3-KNO3 and nitrite-containing mixtures.	Decomposition, impurities, phase equilibria, and compatibility.	Storage/heat-transfe r cost and operating life, supported by finite-temperature mixture sampling. [34, 36]
Carbonate salts.	Li/Na/K carbonate mixtures.	Electrode reactions, degradation, and interfacial energetics.	Molten-carbonate fuel-cell performance and lifetime. [37]
Molten battery electrolytes and catholytes.	NaI-AlCl3 variants and sodium-metal-halide electrolytes.	Complex-ion speciation, redox, and blocking-film formation.	Cycle life, usable power, and electrolyte/electrode selection. [38]
Low-temperature ionic liquids.	Imidazolium-based chloroaluminate liquids.	Coordination, electrochemical stability, and deposition pathways.	Metal deposition and electrolyte-process design in their operating range. [39]
Deep eutectic solvents.	Choline chloride-urea and related salt/molecular mixtures.	Hydrogen bonding, solvation, selectivity, and decomposition.	Lower-temperature processing and separations; neutral molecular components distinguish this branch from purely ionic liquids. [40]
Molten oxide electrolysis, an adjacent ionic-melt family.	Oxide melts for metal production.	Metal-ion redox and anode/electrolyte interfacial chemistry.	Energy use, anode life, and production economics through a prospective QPU route. [41]

Composition variables include eutectic and off-eutectic ratios, multicomponent blends, process species, inhibitors, and impurities. A lower melting temperature changes the tradeoff among viscosity, corrosion, stability, and electrochemical performance. Select a composition within a specified operating envelope and validate the combined properties.

The computational sequence is classical ensemble generation → selected QPU electronic corrections → classical free-energy and transport estimation → experimental formulation decision. The QPU directly supplies electronic observables; liquid sampling and thermodynamic integration mediate their contribution to salt properties. Neutronics, breeding geometry, reactor flow, and engineering heat transport enter through their respective models.

INL GAIN thermophysical-property project for chloride fuel salts provides a research-budget anchor of $525,000, comprising $400,000 DOE and $125,000 industry funding. This is expenditure on a defined property program, useful for sizing a partner-supported research package. [42]

15. Project selection and funding sequence

Begin with one decision owner, one physical observable, a dated classical baseline, and a validation threshold. Establish the quantum mapping and error budget, then cost the complete workflow and schedule its first paid or deployed decision. Fund each next stage according to the expected decision value of the remaining work.

Decision point	Evidence required	Funding action
Define the commercial question.	Named owner, candidate actions, deadline, and value-capture route.	Budget a bounded feasibility study.
Establish the predictive gap.	Strong classical and experimental comparisons at decision accuracy.	Allocate quantum effort to the unresolved kernel.
Validate the executable model.	Converged physical model, preparation method, full resource estimate, and error budget.	Size the workload to the selected decision.
Validate the prediction.	Blind reference or experimental test with a documented action threshold.	Fund the next prototype or process trial.
Qualify and monetize.	Deployment date, reliability evidence, contractual capture, and updated NPV.	Scale resources with validated demand and cash flow.

By linking shared development to multiple named workloads, we can directly measure the resulting savings. A catalyst kernel enables a scalable route into batteries and pharmaceutical chemistry through a common implementation that respects the unique physical models and validation needs of each domain. Using graphene, hBN, and

benzene as calibration targets ensures precision. The commercial potential peaks when these systems resolve an outstanding product question, meaning that strategic timing is the primary driver of economic return.

We prioritize captured, time-dependent incremental decision value over computational expense alone. EUV, catalysts, batteries, and selected metal-site chemistry lead the strategic portfolio; accessible spectroscopy, salt-property, and interconnect studies provide focused entry projects. Classical quantum-native services, reusable software, and paid research support the funding path as validated QPU contributions mature.

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